Microprocessor & Computer Architecture (μpCA)

UE24CS251B

Session - 5A

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Additional Programs

Program-1 : Convert a given 2 digit hexadecimal number to ASCII CODE and store the result in the memory location X and X+1 using ARM7TDMI – ISA.

Example: $(54)_{10}$ to ASCII code :

1. Mask the LS byte.
2. 54 and 0xF0 gives 50.
3. shift 50 right by 4 times
4. 50 will be 05.
5. Now, 05 && 0x30
6. Results in 0x35.
7. Repeat the same for the LS byte.
8. Mask the MS byte.
9. 54 and 0x0F gives 04.
10. Now, 04 && 0x30
11. Results in 0x34.

```
; ARM7TDMI – ISA code:
.TEXT
  LDR R3, =X
  MOV R1, #0x54
   AND R0, R1, #0xF0
   MOV R0, R0, LSR #4
   ORR R0, R0, #0x30
   STR R0, [R3]
    AND R2, R1, #0x0F
    ORR R2, R2, #0x30
    ADD R3, R3, #4
     STR R2, [R3]
.DATA
X: .WORD 0,0
```

Program-2: Write a program using ARM7TDMI – ISA to find the largest 32 bit element stored in an array A.

```
.DATA
            A: .WORD 10,20,30,40,50

.TEXT
                LDR R1,=A
                 LDR R0, [R1]  ; Initially , set first element as the  largest element
(stored in register R0].

                 MOV R3, #0   ; Count Register

LOOP:    LDR R2, [R1, #4]!   ; load the next element
                CMP R0, R2              ; compare
                BLT  Less                   ; if the previous element is less than next
element then store the largest element in register R0.
        L1:  ADD R3, R3, #1
                CMP R3, #4
                BNE LOOP
                B EXIT

Less:    MOV R0, R2
              B L1
EXIT:    SWI 0X011
```

Program - 3: Convert the following C code to ALP using ARM7TDMI – ISA

C statement: while(save[i] == k)
i +=1;

```
.TEXT
            LDR  R1, =K       ; ADDRESS K
            LDR  R0, [R1]     ; VALUE of K
            LDR  R2, =SAVE    ; SAVE is the starting address of the array
            MOV  R3, #0       ; VALUE of I

  LOOP: ADD R5, R2, R3, LSL #2   ; R5 = SAVE+I*4
            LDR R6, [R5,#0]          ; R6 = [R5]

            CMP R0, R6
             BNE EXIT        ; REPEAT if save[I]==K

              ADD R3, R3, #1
              CMP R3, #5
              BNE LOOP
     EXIT:   SWI 0X011

.DATA
      K: .WORD 5
       I: .WORD 0
SAVE: .WORD 5, 5, 5, 50, 35, 40
```

Program – 4: Consider the following c code snippet. Write the compiled ARM assembly code. Use ARM7TDM – ISA.

```c
int example( int g, int h, int
I, int j)
{ int f;
 f= (g+h)-(i+j);
return f;
}
```

```
.TEXT
LDR R4, =F
MOV R0,  #5   ; G
MOV R1,  #10  ; H
MOV R2,  #1   ; I
MOV R3,  #4   ; J

BL example
STR  R0, [R4]

SWI  0X011
```

```
example:
SUB R13, R13, #12
STR R6, [R13,#8]
STR R5, [R13,#4]
STR R4, [R13,#0]

ADD R5, R0, R1  ; G+H
ADD R6, R2, R3  ; I+j
SUB R4, R5, R6  ;F= (G+H)-(I+J)

MOV R0, R4    ;RETURNS F THROUGH R0.

LDR R4,[R13,#0]
LDR R5,[R13,#4]
LDR R6,[R13,#8]

ADD R13, R13, #12
MOV PC, LR

 .DATA
F: .WORD 0
```

Program – 5: Consider the following c code snippet. Write the compiled ARM assembly code. Use ARM7TDMI – ISA. ; Note: Pass parameters to function on stack.

```
int example( int g, int h, int
I, int j)
{ int f;
 f= (g+h)-(i+j);
return f;
}
```

```
. TEXT
LDR R4, =F
MOV R0,  #5   ; G
MOV R1,  #10  ; H
MOV R2,  #1   ; I
MOV R3,  #4   ; J
;  PASS PARAMETRS TO STACK
SUB R13,  R13,  #16
STR R0,  [R13, #12]
STR R1,  [R13, #8]
STR R2,  [R13, #4]
STR R3,  [R13, #0]
BL example
STR  R0,  [R4]
SWI 0X011
```

```
example:
LDR R5, [R13, #0]    ;  J
LDR R6, [R13, #4]     ;  I
LDR R7, [R13, #8]     ;  H
LDR R8, [R13, #12]  ;  G

ADD R9,  R7,   R8    ;  G+H
ADD R10,  R5,  R6   ;  I+j
SUB R11,  R9,  R10   ; F= (G+H)-
(I+J)

MOV R0,  R11     ; RETURNS F
THROUGH R0.

ADD R13,  R13,  #16
MOV PC,  LR

 . DATA
F:  . WORD 0
```

1. Which instruction is used to mask the higher nibble of a byte in ARM?
   A) ORR
   B) EOR
   C) AND
   D) BIC

2. Which instruction converts a 4-bit hex digit into ASCII in Program-1?
   A) AND
   B) ADD
   C) ORR
   D) EOR

1. Which instruction is used to mask the higher nibble of a byte in ARM?
   A) ORR
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2. Which instruction converts a 4-bit hex digit into ASCII in Program-1?
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3. What is the purpose of the instruction MOV PC, LR at the end of the example function?
A) To copy the Program Counter to the Link Register.
B) To restart the function from the beginning.
C) To return to the calling instruction by restoring the return address into the PC.
D) To exit the program and trigger a software interrupt.

4. How many bytes does .WORD reserve?
A) 1
B) 2
C) 4
D) 8

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4. How many bytes does .WORD reserve?
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C) 4
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THANK YOU

Team MPCA

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